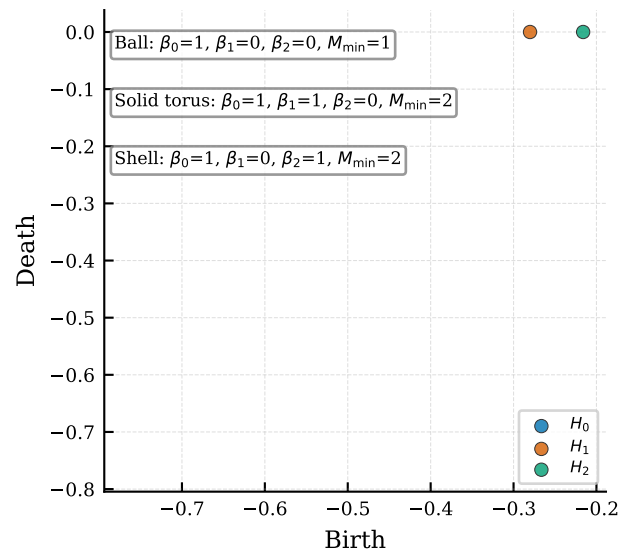
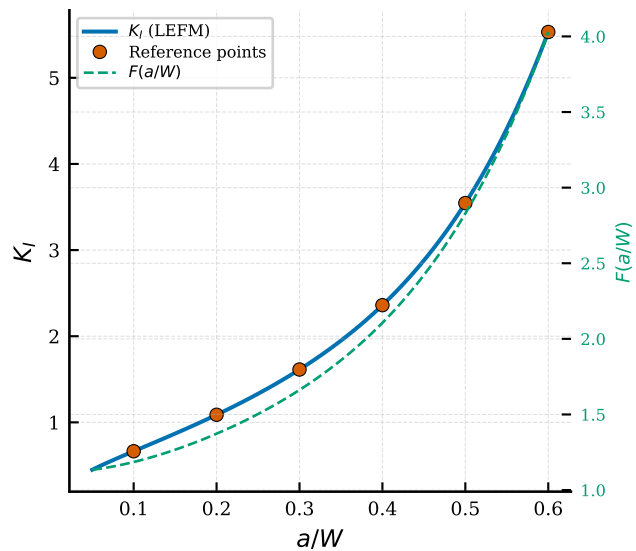
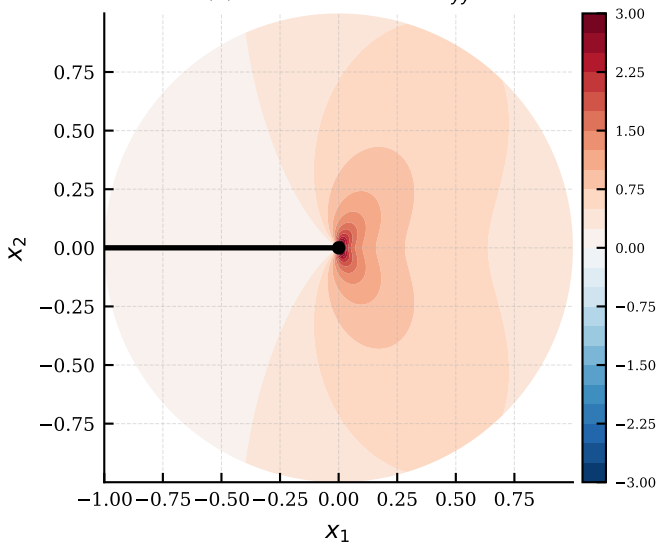


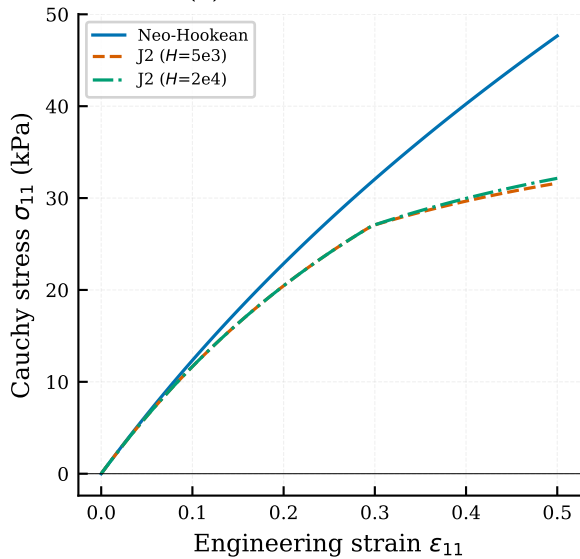
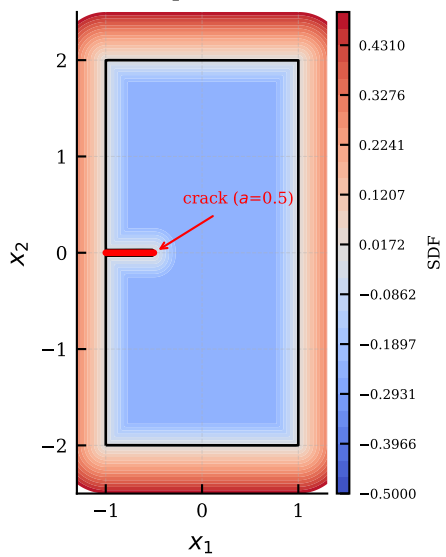
(a) Persistence diagrams



(b) Stress intensity factor

(c) Williams stress  $\sigma_{yy}$ 

(d) Constitutive models

(e) Cracked plate SDF ( $x_3=0$ )

(f) GUDHI persistence timing

